

Advances in Data Structure (Practical) - CSM 105(P), Module – I

Examination 2020

M.Sc., Semester-I in Computer Science

University of Calcutta

You will find your question, which is linked, to your university exam roll number. You are informed to make your file name (in PDF) exactly same as 'your university exam roll number'.

Total marks = 50

1. Sessional + Viva-voce = 25; the scanned handwritten Laboratory Notebooks are to be sent to your college teacher and the answer script of this practical examination is to be sent to 'cucse2020@gmail.com'.

2. Attendance = 00

3. Examination = 25; Duration: 2:00 hours. Scanned answer scripts are to be sent to the same address (cucse2020@gmail.com) within next 15 minutes, after the completion of the exam.

(i) Algorithm in text = 6

(ii) Algorithm at a glance (in pseudo form / step-by- step algorithm / flowchart) = 4

(iii) Data structure(s) = 4

(iv) Algorithmic complexity = 3

(v) Example run = 5

(vi) Special features / Discussions = 3

CSM-105(P)-ADS-115-CSC-201022: State a method to perform insertion operation into an initial empty B-tree of order 5 for the following sequence of key elements: $a_5, g_2, f_1, b_3, k_3, d_6, g_4, m_5, j_2, e_3, s_1, j_1, r_5, z_7, b_6, m_2, m_9, t_3, u_5, q_6$. Instead of computing B-tree after each insertion, compute B* tree after each insertion of the above sequence of keys.

CSM-105(P)-ADS-115-CSC-201023: Consider a sequence S of $n \geq 19$ elements to perform *HeapSort*. Choose appropriate data structure(s) for the said purpose and make clear why and how the data structure(s) is (are) suitable.

CSM-105(P)-ADS-115-CSC-201024: Let X and Y be two lower triangular matrices, each with n rows. Devise a scheme to represent both the triangles in an array Z of size $n \times (n+1)$. Write algorithms to determine the values of

$X[i, j]$ and $Y[i, j]$, $1 \leq i, j \leq n$ from array Z . Choose appropriate data structure(s) for the said purpose, and make clear why and how the data structure(s) is (are) suitable.

CSM-105(P)-ADS-115-CSC-201026: Let A be an $n \times n$ matrix that contains non-zero elements present only in five diagonals, centring and including the major diagonal. If the elements in the band formed by these five diagonals are represented column wise in an array B , with $A[1,1]$ being stored in $B[1]$, obtain an algorithm to determine the value of $A[i, j]$, $1 \leq i, j \leq n$ from array B . Choose appropriate data structure(s) for the said purpose, and make clear why and how the data structure(s) is (are) suitable.

CSM-105(P)-ADS-115-CSC-201027: Consider a sequence S of $n \geq 27$ elements to perform *ShellSort*. Choose appropriate data structure(s) and necessary assumptions for the said purpose and make clear why and how the data structure(s) is (are) suitable.

CSM-105(P)-ADS-115-CSC-201028: Consider a binary search tree and take necessary steps to make it an AVL tree, if needed. Perform deletion of elements from the AVL tree and make the tree AVL after deletion of each element from the tree. Choose appropriate data structure(s) for the said purpose and make clear why and how the data structure(s) is (are) suitable.

CSM-105(P)-ADS-115-CSC-201029: Let P be an $n \times n$ matrix that contains non-zero elements present only in seven diagonals, centring and including the major diagonal. If the elements in the band formed by these seven diagonals are represented row wise in an array Q , with $P[1,1]$ being stored in $Q[1]$, obtain an algorithm to determine the value of $P[i, j]$, $1 \leq i, j \leq n$ from array Q . Choose appropriate data structure(s) for the said purpose, and make clear why and how the data structure(s) is (are) suitable.